

Rational Functions and Their Graphs (Part II)

Rational Function: A function of the form $\frac{A(x)}{B(x)}$, where $A(x)$ and $B(x)$ are polynomials.

Examples: $f(x) = \frac{x}{x^2+6}$, $g(x) = \frac{2x^3+4x^2-2x-1}{x^2-4}$, $h(x) = \frac{-2}{x-1}$

The parent reciprocal function is the most basic type of rational function.

$$f(x) = \frac{1}{x}$$

When graphing rational functions, it is necessary to convert to "proper" form.

Proper Form of a Rational Function: When the degree of the polynomial in the numerator is less than the degree of the polynomial in the denominator.

Examples

Proper

$$\frac{x-5}{x^2-2}, \frac{1}{x^4}, \frac{7x^5-2x+1}{3x^6-4x-2}$$

Improper

$$\frac{x^2-2}{x+6}, \frac{10x^4-2x^3+1}{x^3+3x^2-x+1}, \frac{-8x^7}{x^5+4x}$$

Example: Convert the rational function below to proper form.

$$f(x) = \frac{-7x^4+2x}{x^2+3x-4}$$

$$\begin{array}{r} -7x^2+21x-91 \\ x^2+3x-4 \overline{) -7x^4+2x} \\ \underline{-7x^4-21x^3+28x^2} \\ 21x^3-28x^2+2x \\ \underline{21x^3+63x^2-84x} \\ -91x^2+86x \\ \underline{-91x^2-273x+284} \\ 359x-284 \end{array}$$

$$-7x^2+21x-91 + \frac{359x-284}{x^2+3x-4}$$

Steps for Graphing Rational Functions

- I) Find x and y intercepts and draw them on the graph.
- II) Find asymptotes and draw them as dotted lines.
* The function must be in proper form.
- III) Find the coordinates of holes and draw them on the graph.
- IV) Create a sign chart and use it, along with the previous information, to draw the graph of the function.

The Asymptotes of a Rational Function Theorem

- i) Let $f(x) = \frac{A(x)}{B(x)}$ be a rational function in proper form. If $B(c) = 0$, then $x=c$ is a vertical asymptote of the graph of the function.
- ii) If a rational function is in proper form, then there is a horizontal asymptote on the x-axis.
- iii) If a rational function is in improper form, then there could be a horizontal asymptote and/or an oblique asymptote.

When the Degree of the Numerator Is Equal to the Degree of the Denominator

Graph the following functions. Draw asymptotes as dotted lines and write their equations.

① $f(x) = \frac{4x-5}{x-2}$ Horizontal Asymptote Equation

$$f(x) = 4 + \frac{3}{x-2}$$

$$x-2 \overline{) \begin{array}{r} 4x-5 \\ 4x-8 \\ \hline 3 \end{array}}$$

$$\boxed{y=4}$$

Vertical Asymptote Equation(s)

$$\boxed{x=2}$$

Y-Intercept

$$f(0) = \frac{4(0)-5}{0-2} = \frac{5}{2}$$

$$\boxed{(0, 2\frac{1}{2})}$$

X-Intercept

$$f(x) = 0$$

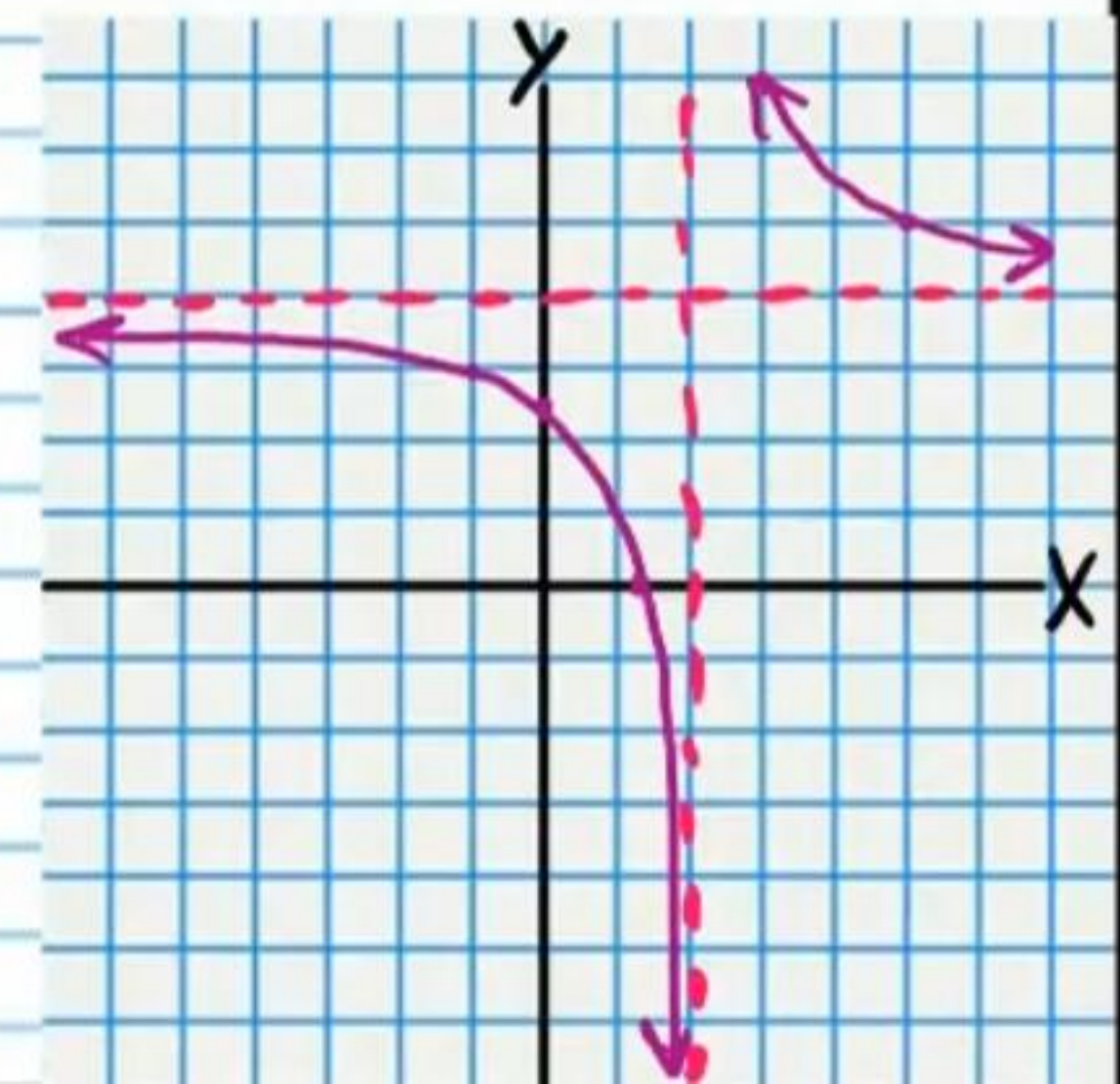
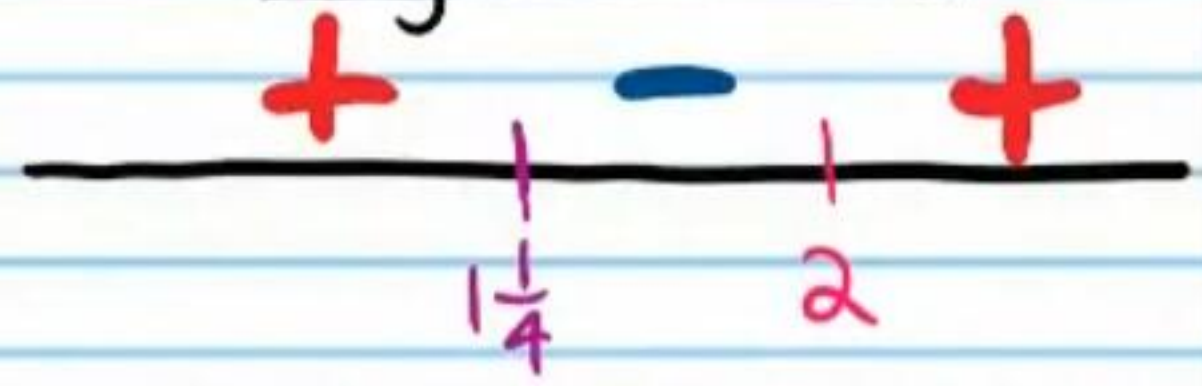
$$\frac{4x-5}{x-2} = 0$$

$$4x-5 = 0$$

$$x = 1\frac{1}{4}$$

$$\boxed{(1\frac{1}{4}, 0)}$$

Sign Chart



② $f(x) = \frac{x^2+6x+5}{-x^2+2x+8} = \frac{(x+5)(x+1)}{-(x^2-2x-8)} = \frac{(x+5)(x+1)}{-(x-4)(x+2)}$

Horizontal Asymptote Equation

$$f(x) = -1 + \frac{8x+13}{-x^2+2x+8}$$

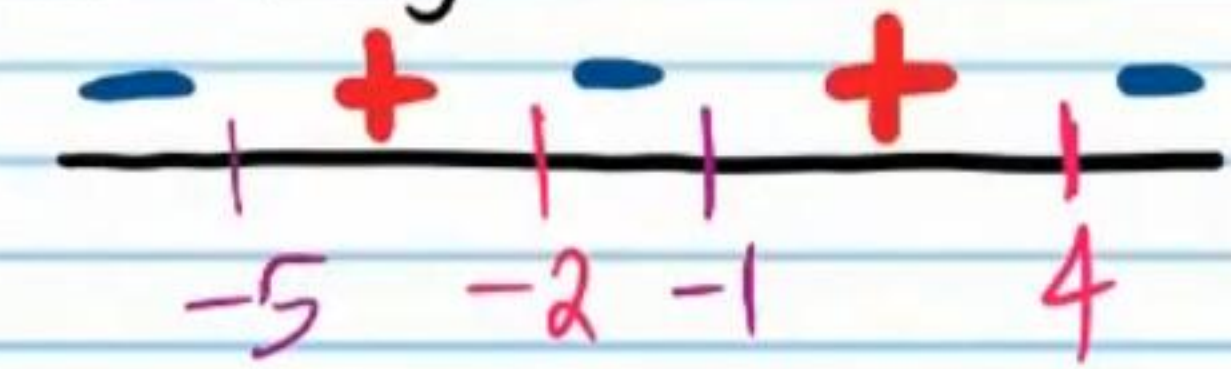
$$-x^2+2x+8 \overline{) \begin{array}{r} x^2+6x+5 \\ \hline x^2-2x-8 \\ \hline 8x+13 \end{array}}$$

$$\boxed{y=-1}$$

Vertical Asymptote Equation(s)

$$\boxed{x=4, x=-2}$$

Sign Chart



Y-Intercept

$$f(0) = \frac{(0+5)(0+1)}{-(0-4)(0+2)} = \frac{5}{8}$$

$$\boxed{(0, \frac{5}{8})}$$

X-Intercept

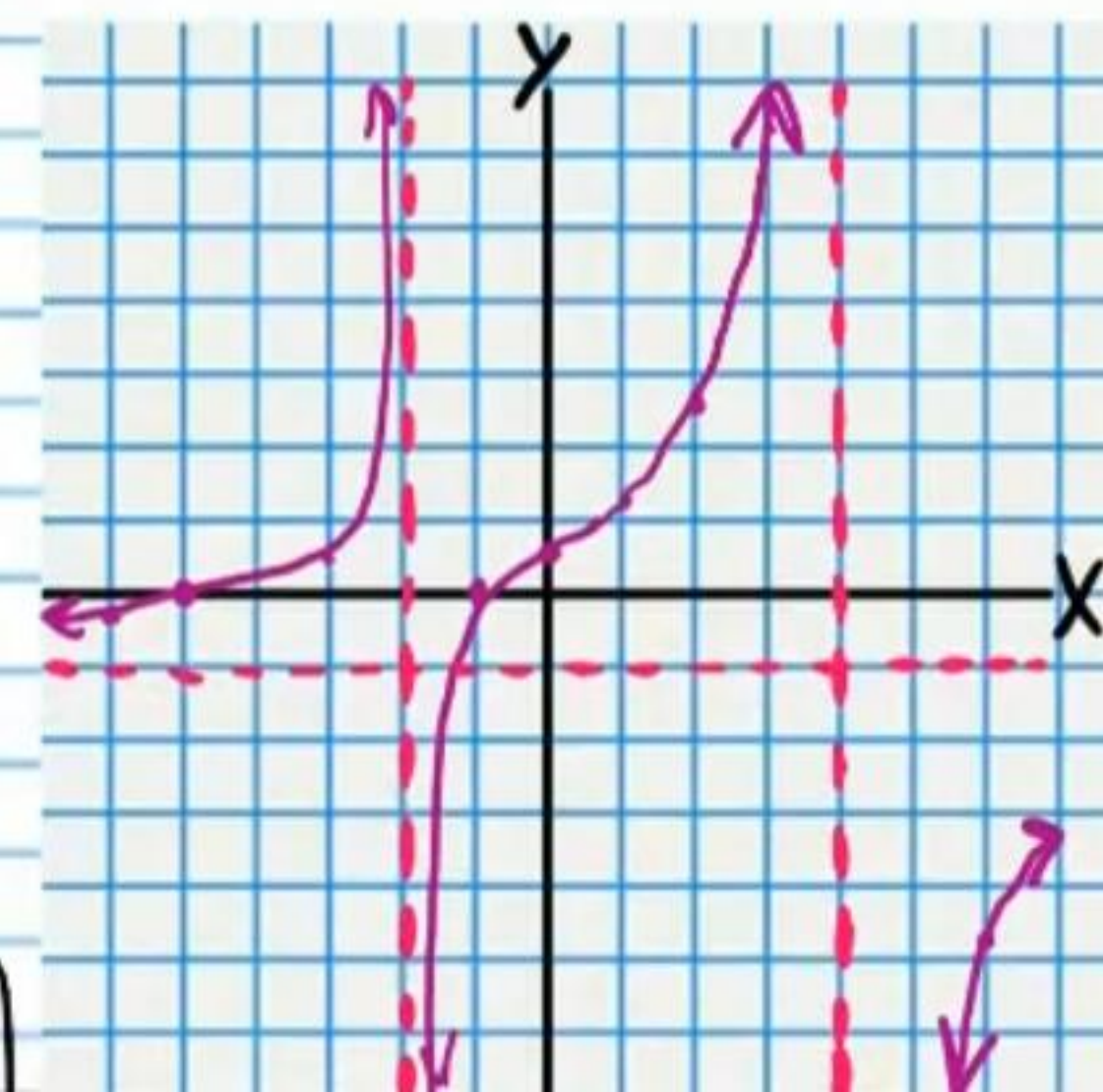
$$f(x) = 0$$

$$\frac{(x+5)(x+1)}{-(x-4)(x+2)} = 0$$

$$x = -5$$

$$x = -1$$

$$\boxed{(-5, 0) \text{ \& } (-1, 0)}$$



③ $f(x) = \frac{-3x+1}{x+1}$ Horizontal Asymptote Equations

$$f(x) = -3 + \frac{4}{x+1}$$

$$x+1 \overline{) -3x+1}$$

$$\underline{-3x-3}$$

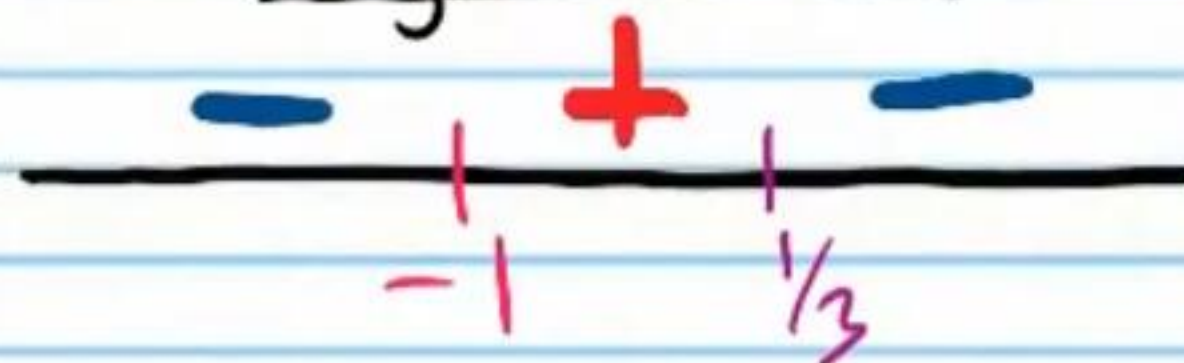
$$4$$

$$\boxed{y = -3}$$

Vertical Asymptote Equation(s)

$$\boxed{x = -1}$$

Sign Chart



Y-Intercept

$$f(0) = \frac{-3(0)+1}{0+1} = 1$$

$$\boxed{(0, 1)}$$

X-Intercept

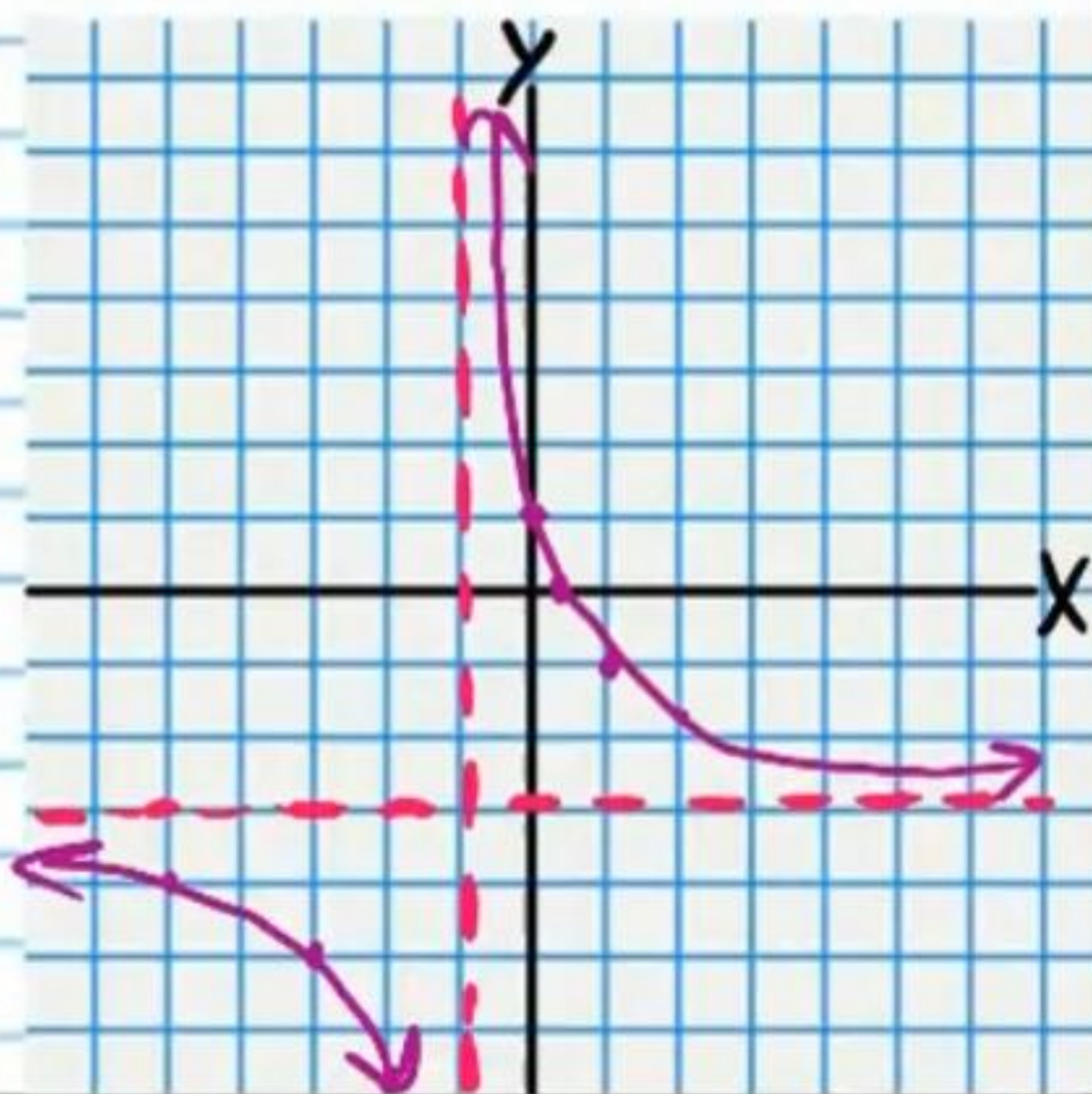
$$f(x) = 0$$

$$\frac{-3x+1}{x+1} = 0$$

$$\rightarrow -3x+1=0$$

$$x = \frac{1}{3}$$

$$\boxed{(\frac{1}{3}, 0)}$$



④ $f(x) = \frac{2x^2-3x-2}{x^2+2x-3} = \frac{(2x+1)(x-2)}{(x+3)(x-1)}$

Horizontal Asymptote Equation

$$f(x) = 2 + \frac{-7x+4}{x^2+2x-3}$$

$$\boxed{y = 2}$$

$$x^2+2x-3 \overline{) 2x^2-3x-2}$$

$$\underline{2x^2+4x-6}$$

$$-7x+4$$

Vertical Asymptote Equation(s)

$$\boxed{x = -3, x = 1}$$

Y-Intercept

$$f(0) = \frac{2(0)^2-3(0)-2}{0^2+2(0)-3}$$

$$= \frac{2}{3} \quad \boxed{(0, \frac{2}{3})}$$

X-Intercept

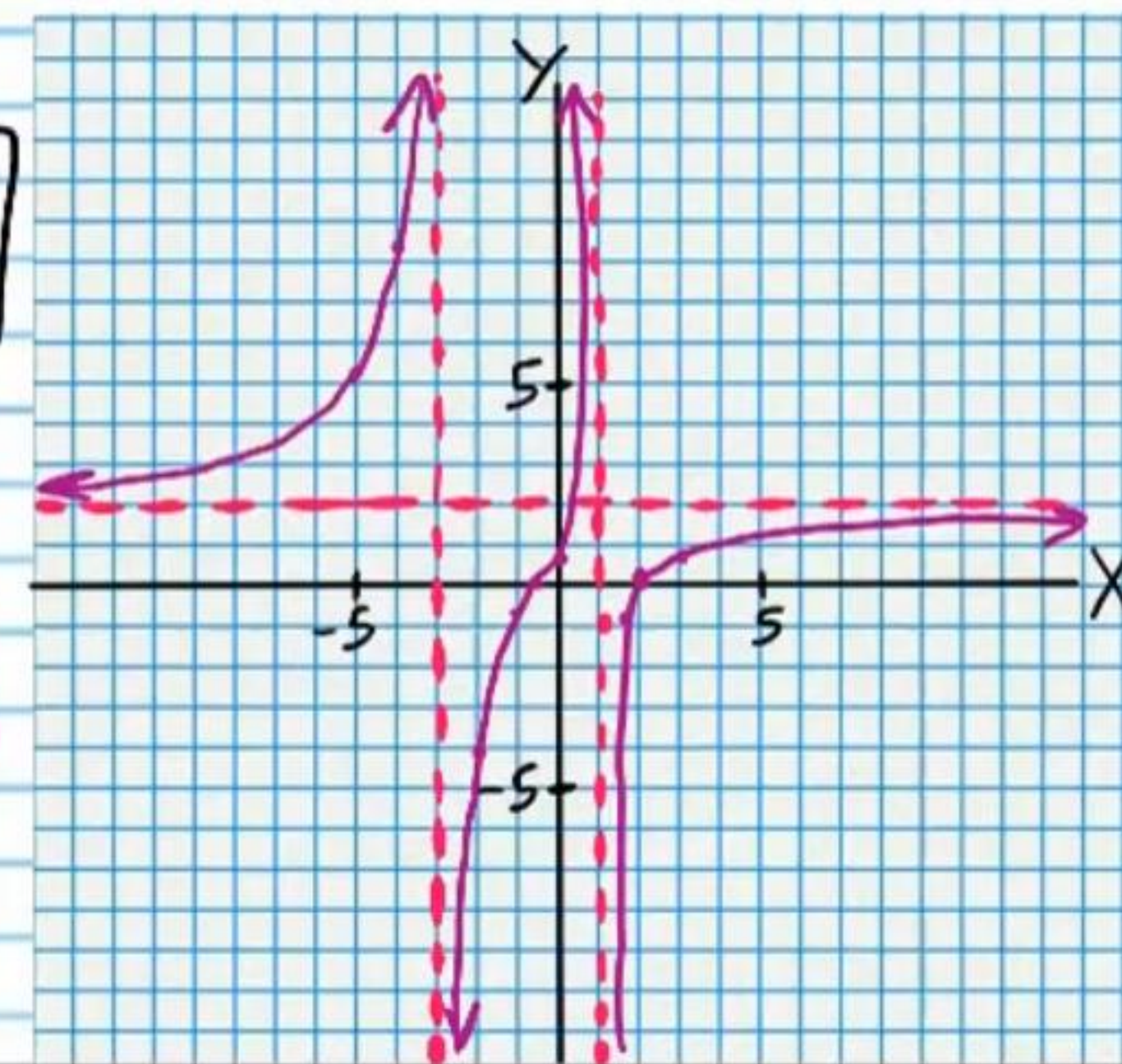
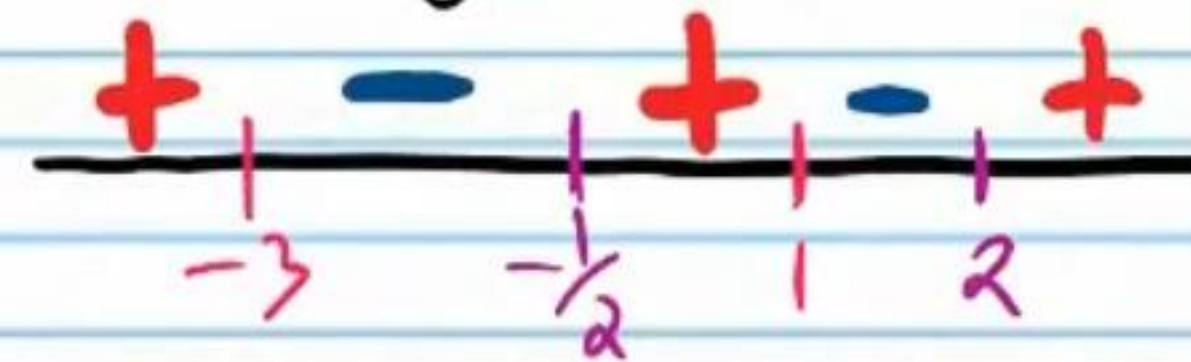
$$f(x) = 0$$

$$\frac{(2x+1)(x-2)}{(x+3)(x-1)} = 0 \quad \begin{matrix} -13(-9) \\ -4(-8) \end{matrix}$$

$$2x+1=0 \text{ or } x-2=0$$

$$\boxed{(-\frac{1}{2}, 0) \text{ or } (2, 0)}$$

Sign Chart



When the Degree of the Numerator Is Greater than the Degree of the Denominator

$$⑤ f(x) = \frac{x^4 - 16}{x^3 + 3x^2 - 4x - 12} = \frac{(x^2 - 4)(x^2 + 4)}{x^2(x+3) - 4(x+3)} = \frac{(x-2)(x+2)(x^2+4)}{(x+3)(x^2-4)}$$

Oblique Asymptote Equation

$$g(x) = \frac{x^2+4}{x+3} = \frac{x-3}{x+3} \cdot \frac{x^2+4}{x^2+3x} = \frac{(x-2)(x+2)(x^2+4)}{(x+3)(x-2)(x+2)}$$

$$g(x) = x-3 + \frac{13}{x+3}$$

Holes: $x=2, x=-2$

Oblique Asymptote Equation: $y = x-3$

Vertical Asymptote Equation(s)

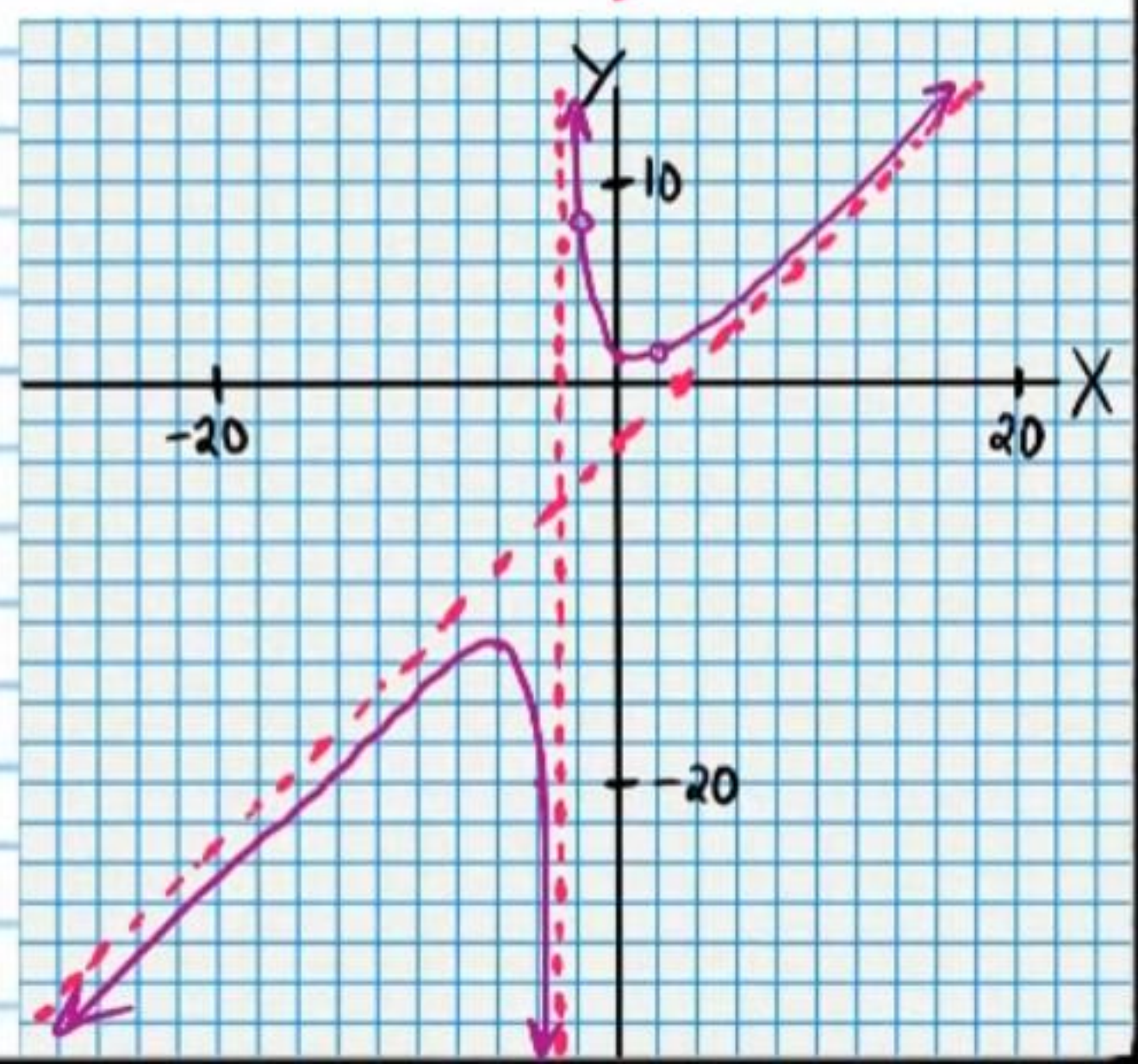
Vertical Asymptote Equation(s): $x = -3$

Y-Intercept: $(0, \frac{4}{3})$

X-Intercept

$$x^2 + 4 = 0$$

$$x = \pm 2i \text{ None}$$



$$⑥ f(x) = \frac{-x^2 + x + 2}{x-3} = \frac{-(x^2 - x - 2)}{x-3} = \frac{-(x-2)(x+1)}{x-3}$$

Vertical Asymptote Equation(s)

Vertical Asymptote Equation(s): $x = 3$

Oblique Asymptote Equation

$$f(x) = -x-2 - \frac{4}{x-3}$$

Oblique Asymptote Equation: $y = -x-2$

Y-Intercept

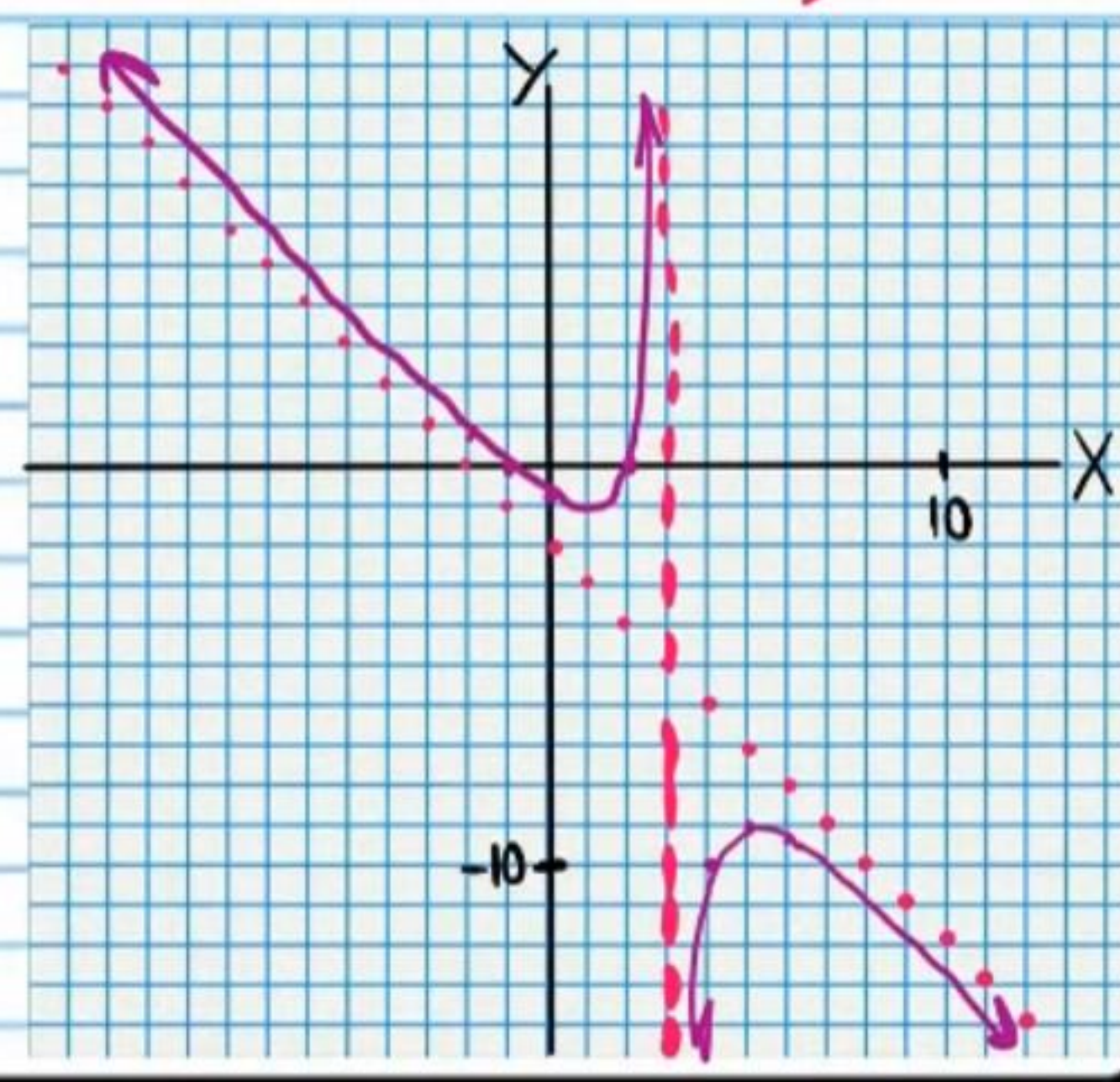
$$f(0) = -\frac{2}{3}$$

Y-Intercept: $(0, -\frac{2}{3})$

X-Intercept

X-Intercept: $(2, 0) \text{ \& } (-1, 0)$

$$\begin{array}{r} -x-2 \\ x-3 \overline{) -x^2+x+2} \\ \underline{-x^2+3x} \\ -2x+2 \\ \underline{-2x+6} \\ -4 \end{array}$$



$$\textcircled{7} f(x) = \frac{x^4 - 81}{x^3 - 2x^2 - 9x + 18} = \frac{(x^2 - 9)(x^2 + 9)}{x^2(x-2) - 9(x-2)} = \frac{(x+3)(x-3)(x^2+9)}{(x-2)(x^2-9)}$$

Oblique Asymptote Equation $= \frac{(x+3)(x-3)(x^2+9)}{(x-2)(x-3)(x+3)}$

$$g(x) = \frac{x^2+9}{x-2}$$

$$x-2 \overline{) \begin{array}{r} x+2 \\ x^2+9 \\ \hline x^2-2x \end{array}}$$

$$g(x) = x+2 + \frac{13}{x-2}$$

$$\begin{array}{r} 2x+9 \\ 2x-4 \\ \hline 13 \end{array}$$

Hole(s)

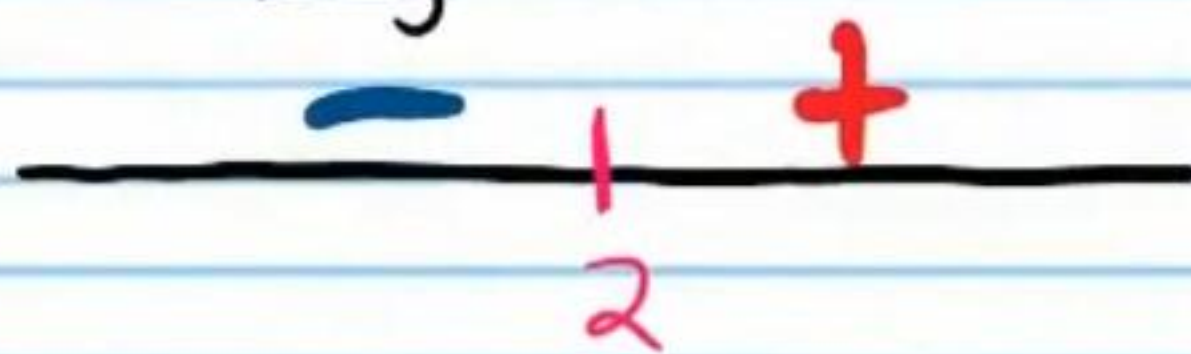
$$x=3, x=-3$$

$$\boxed{y = x+2}$$

Vertical Asymptote Equation(s)

$$\boxed{x = 2}$$

Sign Chart



Y-Intercept

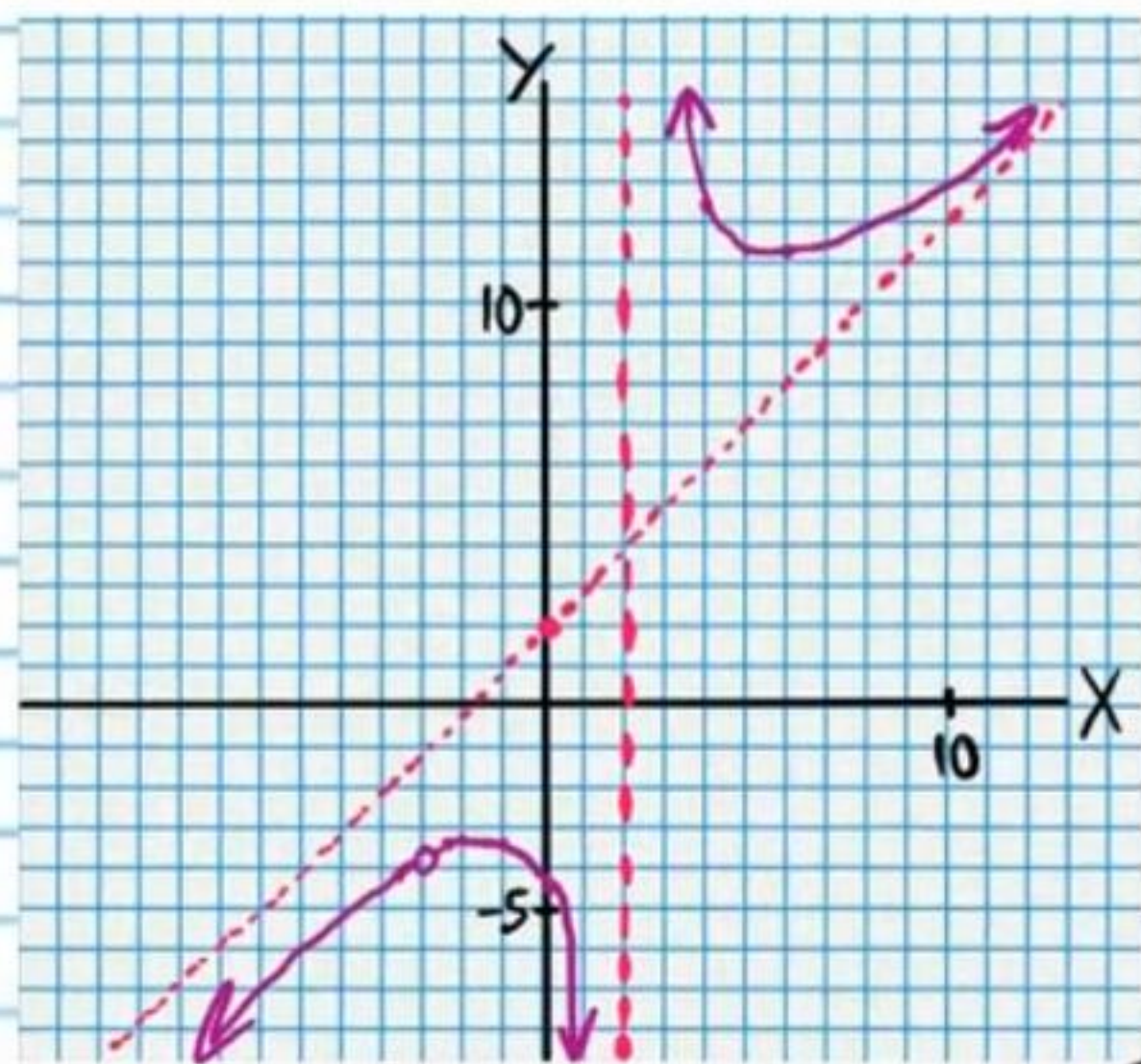
$$f(0) = -\frac{81}{18} = -\frac{9}{2}$$

$$\boxed{(0, -4\frac{1}{2})}$$

X-Intercept

$$x^2+9=0$$

$$x = \pm 3i$$



$$\textcircled{8} f(x) = \frac{x^2+x-6}{-x-1} = \frac{(x-2)(x+3)}{-1(x+1)}$$

Oblique Asymptote Equation

$$f(x) = -x - \frac{6}{-x-1} \quad -x-1 \overline{) \begin{array}{r} -x \\ x^2+x-6 \\ \hline x^2+x \\ \hline -6 \end{array}}$$

$$f(x) = -x + \frac{6}{x+1}$$

$$\boxed{y = -x}$$

Vertical Asymptote Equation(s)

$$\boxed{x = -1}$$

Y-Intercept

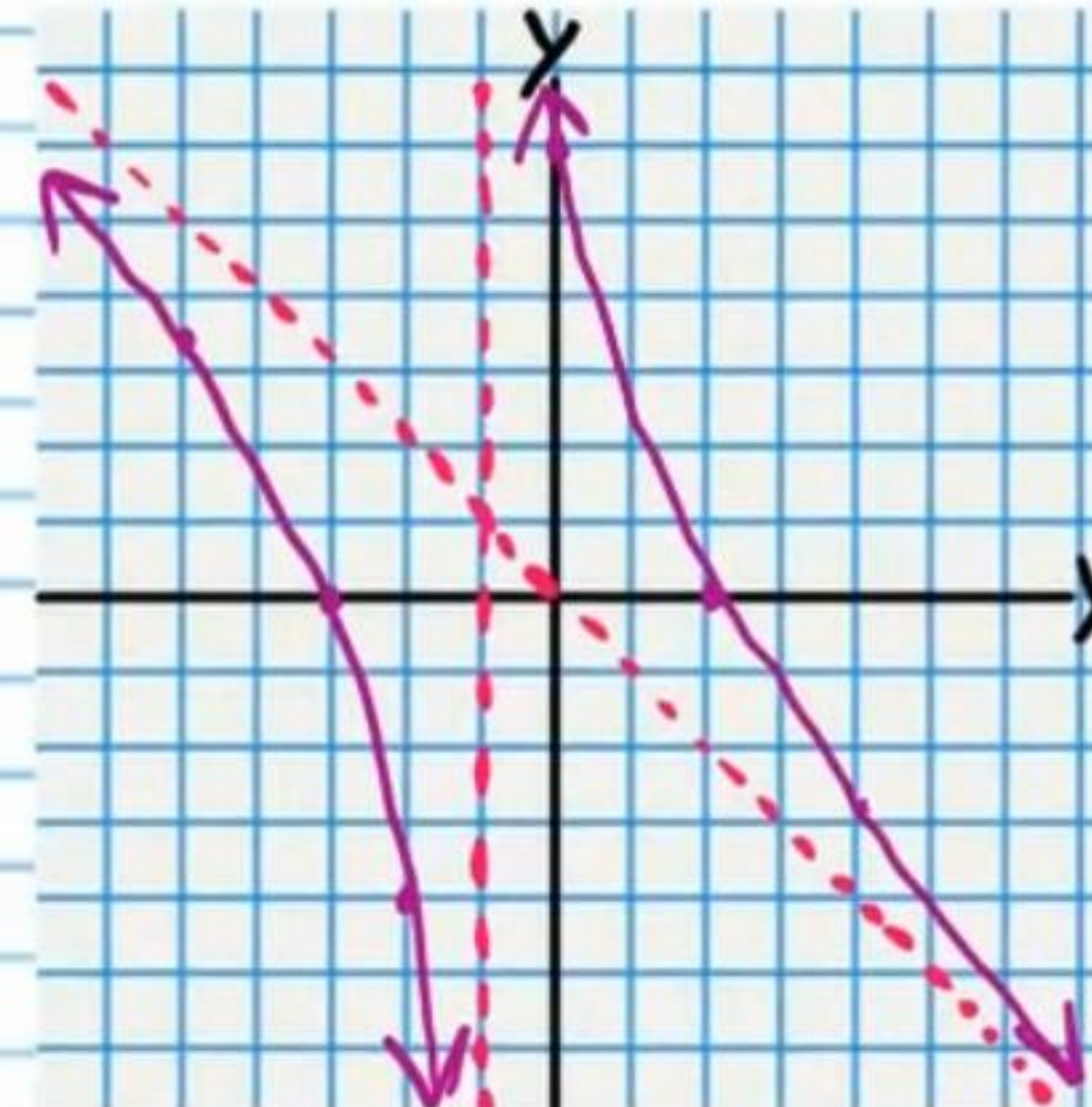
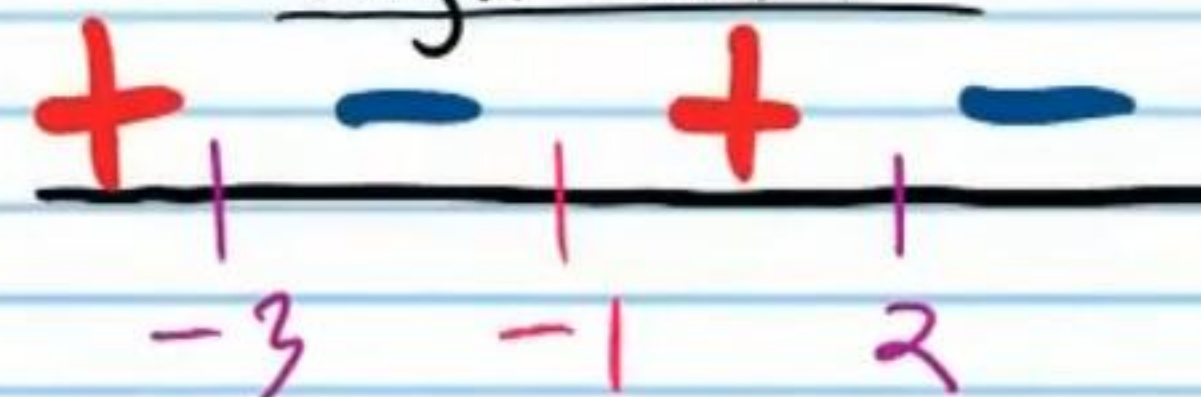
$$f(0) = 6$$

$$\boxed{(0, 6)}$$

X-Intercept

$$\boxed{x = 2, x = -3}$$

Sign Chart



For each function, find the information requested.
It is not necessary to graph each function.

$$⑨ f(x) = \frac{x^5 + x^4}{x^3 + 4x^2 + 3x}$$

$$= \frac{x^4(x+1)}{x(x^2+4x+3)}$$

$$= \frac{x \cdot x \cdot x \cdot x (x+1)}{x (x+3) (x+1)}$$

Asymptote Curve

$$g(x) = \frac{x^3}{x+3}$$

$$g(x) = x^2 - 3x + 9 - \frac{27}{x+3}$$

$$\boxed{y = x^2 - 3x + 9}$$

Y-Intercept

None

Hole(s)

$$\boxed{x = -1}$$

$$\boxed{x = 0}$$

Vertical Asymptote Equation(s)

$$\boxed{x = -3}$$

$$\begin{array}{r} x^2 - 3x + 9 \\ x+3 \overline{) x^3} \\ \underline{x^3 + 3x^2} \\ -3x^2 \\ \underline{-3x^2 - 9x} \\ 9x \\ \underline{9x + 27} \\ -27 \end{array}$$

X-Intercept

None

$$⑩ f(x) = \frac{x^3 - 2x^2 + x - 2}{x+2}$$

$$\frac{x^2(x-2) + 1(x-2)}{x+2}$$

$$\frac{(x-2)(x^2+1)}{x+2}$$

Y-Intercept

$$f(0) = -1 \quad \boxed{(0, -1)}$$

X-Intercept

$$\boxed{(2, 0)}$$

Vertical Asymptote Equation(s)

$$\boxed{x = -2}$$

Asymptote Curve

$$f(x) = x^2 - 4x + 9 - \frac{20}{x+2}$$

$$\boxed{y = x^2 - 4x + 9}$$

$$\begin{array}{r} x^2 - 4x + 9 \\ x+2 \overline{) x^3 - 2x^2 + x - 2} \\ \underline{x^3 + 2x^2} \\ -4x^2 + x - 2 \\ \underline{-4x^2 - 8x} \\ 9x - 2 \\ \underline{9x + 18} \\ -20 \end{array}$$

$$\textcircled{11} f(x) = \frac{x^4 + 2x^2}{x^2 - 5x} = \frac{x^2(x^2 + 2)}{x(x-5)} = \frac{x \cdot \cancel{x} (x^2 + 2)}{\cancel{x} (x-5)}$$

Asymptote Curve

$$g(x) = \frac{x(x^2 + 2)}{x-5}$$

Hole(s)

$$\boxed{x=0}$$

$$g(x) = \frac{x^3 + 2x}{x-5}$$

Y-Intercept

None

$$x-5 \overline{) \begin{array}{r} x^2 + 5x + 27 \\ x^3 + 2x \\ \hline x^3 - 5x^2 \end{array}}$$

X-Intercept

None

$$\begin{array}{r} 5x^2 + 2x \\ 5x^2 - 25x \\ \hline 27x \end{array}$$

$$\begin{array}{r} 27x \\ 27x - 135 \\ \hline 135 \end{array}$$

$$g(x) = x^2 + 5x + 27 + \frac{135}{x-5}$$

$$\boxed{y = x^2 + 5x + 27}$$

Vertical Asymptote Equation(s)

$$\boxed{x=5}$$

$$\textcircled{12} f(x) = \frac{x^3 + 4x^2 - x - 4}{x-2} = \frac{x^2(x+4) - (x+4)}{x-2} = \frac{(x+4)(x^2-1)}{x-2}$$

Asymptote Curve

$$= \frac{(x+4)(x-1)(x+1)}{\cancel{x-2}}$$

$$x-2 \overline{) \begin{array}{r} x^2 + 6x + 11 \\ x^3 + 4x^2 - x - 4 \\ \hline x^3 - 2x^2 \end{array}}$$

Y-Intercept

$$\boxed{(0, 2)}$$

$$\begin{array}{r} 6x^2 - x - 4 \\ 6x^2 - 12x \\ \hline 11x - 4 \end{array}$$

$$f(x) = x^2 + 6x + 11 + \frac{18}{x-2}$$

$$\begin{array}{r} 11x - 4 \\ 11x - 22 \\ \hline 18 \end{array}$$

$$\boxed{y = x^2 + 6x + 11}$$

Vertical Asymptote Equation(s)

$$\boxed{x=2}$$

X-Intercept

$$\boxed{(-1, 0), (1, 0), (-4, 0)}$$